

Homework (Carolina Reaper)

- Q1.** A music DJ uses a relation to map song genres to dance styles. If the relation is $\{(rock, salsa), (pop, waltz), (rock, tango)\}$, is this a function?
- (A) Yes
(B) No
(C) Maybe
(D) Only for rock
(E) Only for pop
- Q2.** A painter uses a function $f(x) = 2x$ to calculate the frame size for a canvas of width x . What is $f(5)$?
- (A) 5
(B) 10
(C) 12
(D) 15
(E) 20
- Q3.** The frequency of a guitar string is given by the function $f(x) = 440 \cdot 2^{\frac{x-4}{12}}$. What is the frequency when $x = 6$?
- (A) 440
(B) 660
(C) 880
(D) 1100
(E) 1320
- Q4.** A dance instructor uses a relation to map rhythm patterns to tempo. If the graph of the relation has multiple y -values for $x = 120$, is this a function?
- (A) Yes
(B) No
(C) Maybe
(D) Only for $x = 120$
(E) Only for $y > 100$
- Q5.** A music theorist uses the function $f(x) = x^2$ to model the resonance of a sound wave. What is the domain of this function?
- (A) $(-\infty, \infty)$
(B) $(-10, 10)$
(C) $[0, 10]$
(D) $(-5, 5]$
(E) $[0, \infty)$
- Q6.** A sound engineer uses the relation $\{(0, 0), (1, 2), (2, 4), (3, 6)\}$ to map audio levels to decibel values. What is the range of this relation?
- (A) $\{0, 2, 4, 6\}$
(B) $\{0, 1, 2, 3\}$
(C) $\{2, 4, 6, 8\}$
(D) $\{0, 1, 4, 9\}$
(E) $\{0, 2, 4, 8\}$
- Q7.** A composer uses the function $f(x) = 3x - 2$ to determine the length of a musical phrase. What is $f(8)$?
- (A) 20
(B) 22
(C) 24
(D) 26
(E) 28
- Q8.** A musicologist uses a relation to map historical periods to musical styles. If the relation is a function, which statement is true?
- (A) Each historical period maps to one musical style
(B) Each musical style maps to one historical period
(C) Each historical period maps to multiple musical styles
(D) Each musical style maps to multiple historical periods
(E) The relation is not a function

Open Ended Questions (FRQ)

- Q9.** Suppose $f(x) = x^3 - 2x^2 + x - 1$ is a function modeling the volume of a musical instrument. Determine the domain and range of this function and provide a brief explanation.
- Q10.** The graph of a relation R is given by $y = |x - 2| + 1$. Determine whether R is a function and explain why or why not. If it is a function, evaluate $f(0)$ and $f(4)$.

Chef's Tips

- Tip 1: Check for any duplicate questions or answers
- Tip 2: Ensure all questions are relevant to the unit and topic
- Tip 3: Review the test for any ambiguities or unclear language